

Analiza porównawcza ryzyka w modelach GARCH

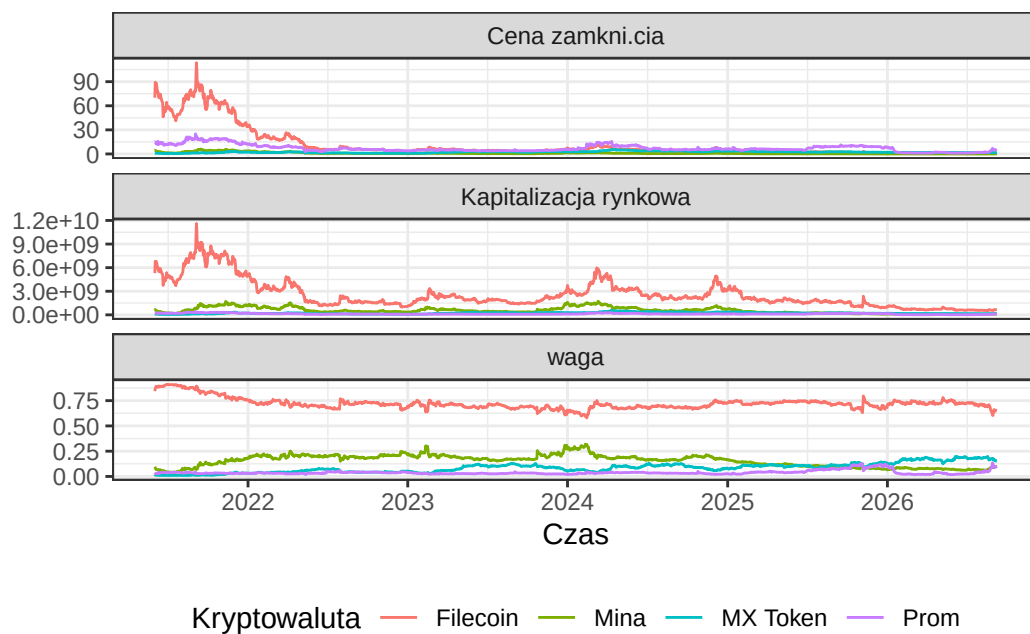
Filip Piotrowski

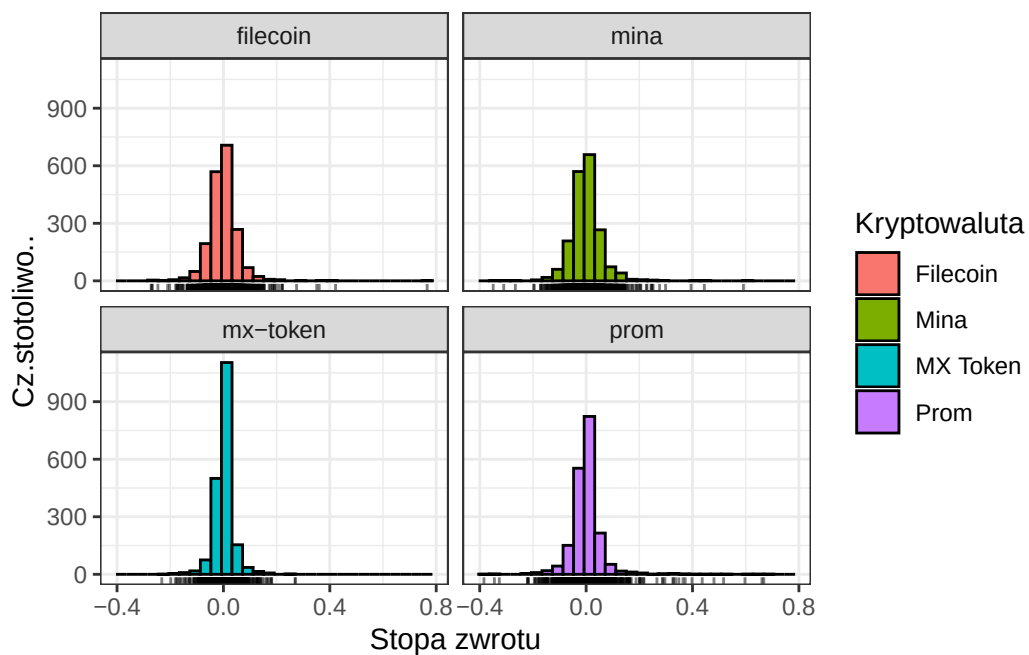
Marcel Marszałek

2026-09-10

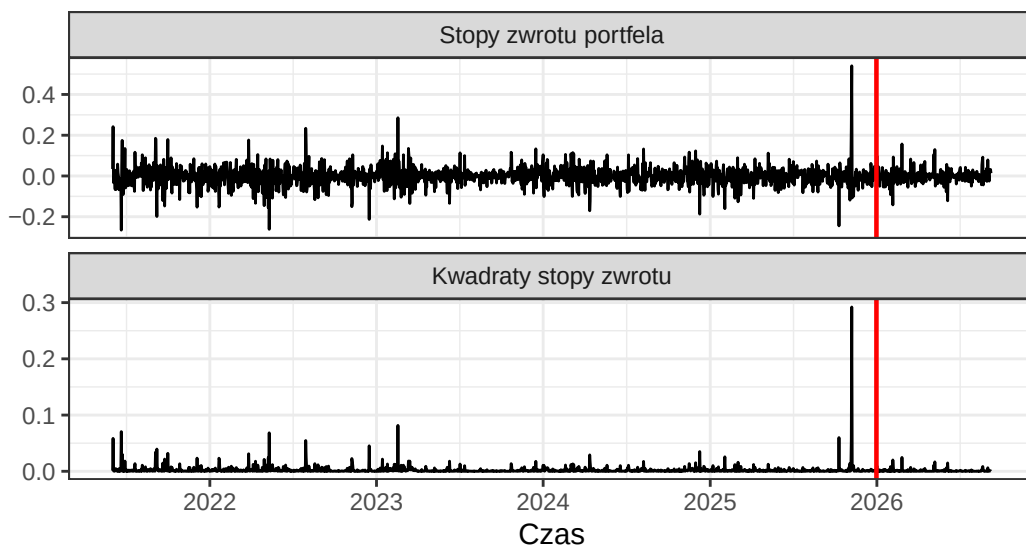
Zbiór danych

Eksploracyjna analiza danych





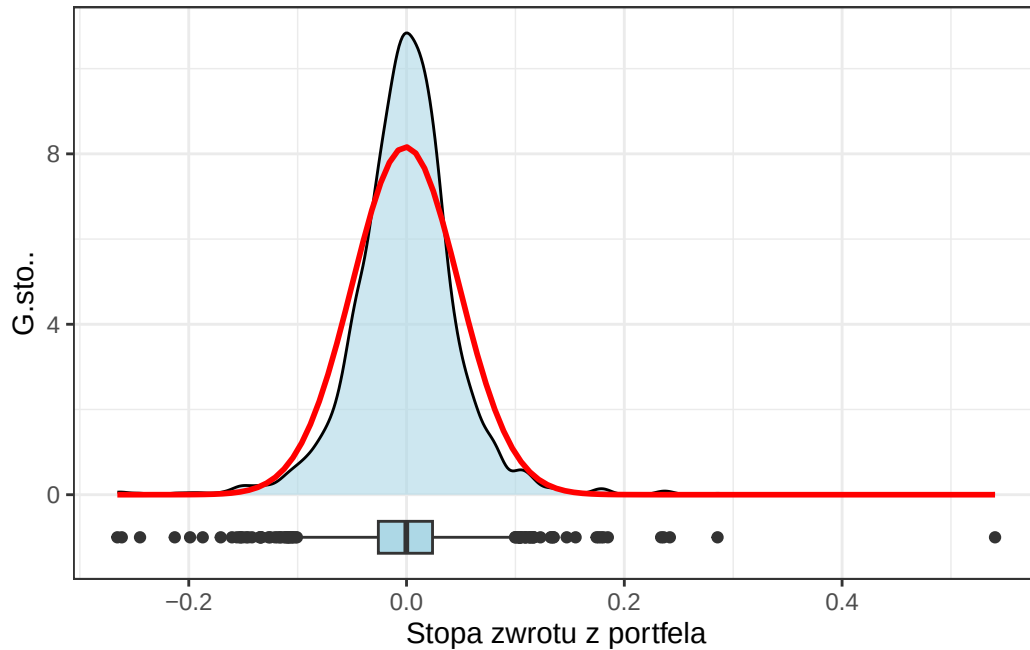
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-0.2655978	-0.0253703	-0.0004562	-0.0008973	0.0224418	0.5404989



| Podzia. train/test — Stopy zwrotu

Wstępna diagnostyka

Rozkład zmiennej modelowanej



Jarque Bera Test

```
data: r_train  
X-squared = 11106, df = 2, p-value < 2.2e-16
```

Stacjonarność

Augmented Dickey-Fuller Test

```
data: r_test  
Dickey-Fuller = -5.9093, Lag order = 6, p-value = 0.01  
alternative hypothesis: stationary
```

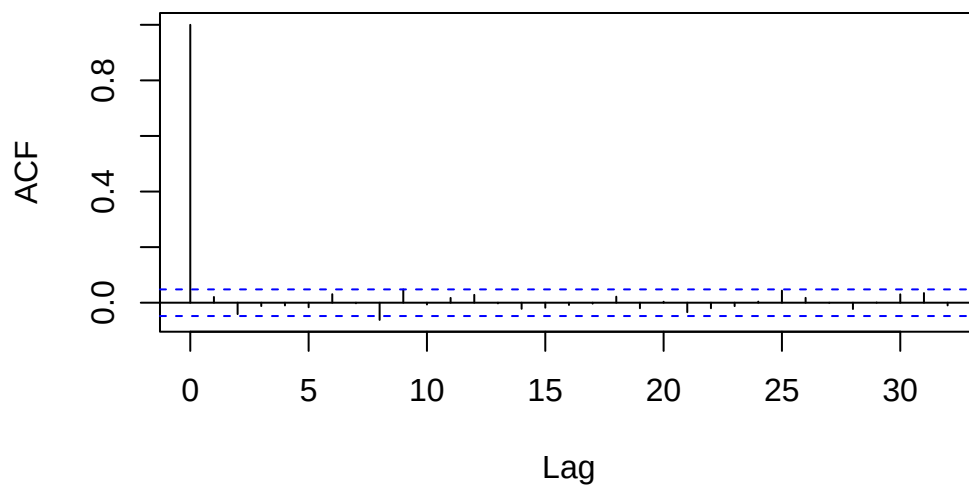
KPSS Test for Level Stationarity

data: r_train

KPSS Level = 0.086451, Truncation lag parameter = 8, p-value = 0.1

Autokorelacja

ACF stóp zwrotu portfela



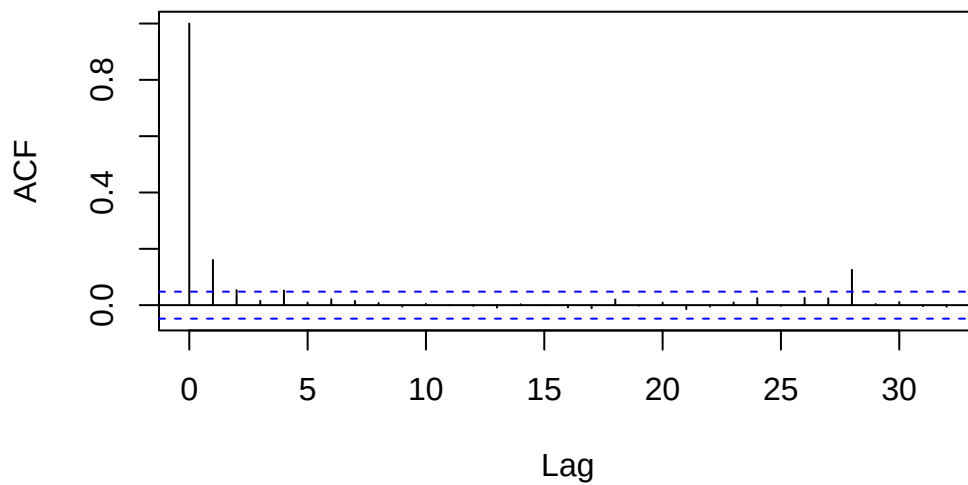
Box-Ljung test

data: r_train

X-squared = 16.351, df = 10, p-value = 0.09001

Warunkowa heteroskedastyczność

ACF kwadratów stóp zwrotu



Box-Ljung test

```
data: r_train^2  
X-squared = 54.338, df = 10, p-value = 4.197e-08
```

ARCH LM-test; Null hypothesis: no ARCH effects

```
data: r_train  
Chi-squared = 51.681, df = 10, p-value = 1.307e-07
```